Exercise path

Start with exercises 1 and 2. Then choose one of exercises 3 or 4 and do it without help. Exercises 5 to 7 are a reserve for transfer, investigation or programming: the teacher designates one. Exercise 8 is revisited after a delay.

Exercise 6.1 · A collection of derivatives

Consider the functions

$$f(x) = 3x^4 - 2x + 7, \quad g(x) = (x^2 + 2)(x - 1), \quad h(x) = \frac{2x + 1}{x + 3}.$$

1. State the domain of definition of each of these functions.
2. Calculate their derivatives, naming the rule used each time. For g, first keep the two terms of the product rule; for h, write the numerator before simplifying it.
3. Check the result for g by expanding its expression before differentiating.

Exercise 6.2 · A complete table

The function f is defined on $[0, 4]$ by $f(x) = x^3 - 6x^2 + 9x + 1$. We look for its minimum and maximum values on this interval.

1. Calculate and factorise $f'(x)$, then study its sign on $[0, 4]$.
2. Draw the monotonicity table of f, giving the values at the endpoints and at the points where the derivative vanishes.
3. Give the minimum and the maximum of f on the interval, as well as all the x-coordinates where they are reached.

Exercise 6.3 · A quotient

For every strictly positive real number x, let $f(x) = x + \dfrac{4}{x}$.

1. Study the monotonicity of f on $(0, +\infty)$. Present the factors that determine the sign of the derivative.
2. Deduce the smallest value of $x + \dfrac{4}{x}$, then write an inequality valid for all $x > 0$. State exactly its equality case.

Exercise 6.4 · Comparing two curves

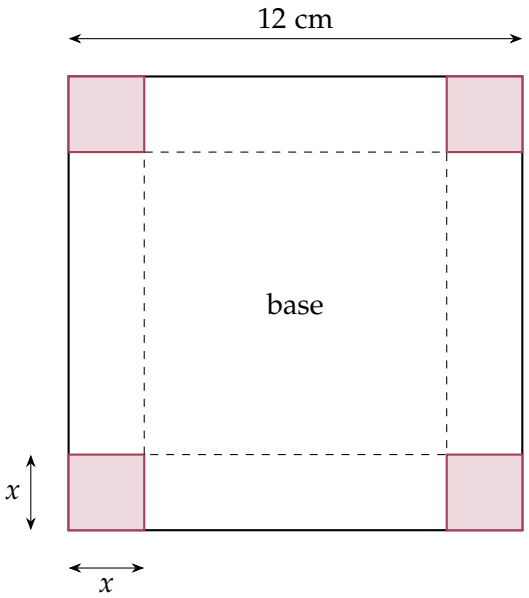
The functions $f(x) = x^3$ and $g(x) = x$ are defined on $\mathbb{R}$. Let $\mathcal{C}_f$ and $\mathcal{C}_g$ be their curves in the same frame.

1. Determine the points of intersection of these two curves.
2. Study their relative position on $\mathbb{R}$ using a suitable method. Express your result as the intervals where $\mathcal{C}_f$ is above or below $\mathcal{C}_g$.

Exercise 6.5 · A box

A box without a lid is made from a square sheet of side 12 cm. A square of side x cm is cut from each of the four corners, then the remaining edges are folded up. The thickness of the sheet is neglected.

1. Draw the net and express the height and the two dimensions of the base of the box. State the values of x that give a box of nonzero volume.
2. Express the volume $V(x)$ in cubic centimetres. Determine the dimensions that make this volume maximal.
3. Check the values at the two endpoints of the domain and give the exact maximum volume.



The four coloured regions are removed; the dotted lines show the folds. The drawing uses an illustrative value of x and does not give the optimum.

Exercise 6.6 · Stationary without an extremum

A student claims: "If the derivative vanishes at a point, the function necessarily has a local extremum at that point." We study the function $f(x) = x^3$ to test this claim.

1. Calculate $f'(0)$.
2. Compare with $f(0)$ the values of $f(x)$ for negative then positive real numbers x, as close to 0 as we like.
3. Conclude about the existence of a local extremum at 0 and about the claim proposed.

Exercise 6.7 · Comparing strategies

We wish to prove that, for every real number x, $x^2 - 6x + 12 \geq 3$.

1. Give a proof by rewriting the expression using a square.
2. Give a second proof by studying the monotonicity of a function.
3. State the equality case. Compare the two methods: which requires the least calculation here?

Exercise 6.8 · Delayed review

The reciprocal function $f(x) = \dfrac{1}{x}$ has domain $D = \mathbb{R} \setminus \{0\}$. A student calculates $f'(x) = -\dfrac{1}{x^2} < 0$ and concludes that f is decreasing on all of D.

1. Identify the missing hypothesis when applying the theorem that links the sign of the derivative to monotonicity.
2. State the two intervals on which the conclusion is correct.
3. Using the numbers -1 and 1, construct a counterexample to the conclusion on all of D.

Homework. Do only the task designated at the end of the session, in the dated progression: 10–20 min at most. The exercises depend on the topics already studied, so some questions are deferred. Revisit exercise 8 one week after the chapter (10 min). If the allotted time is exceeded, note the difficulty and bring your attempts; do not pile up the whole reserve.

To do in the CQFD textbook

These exercises are done in your textbook. The progression indicates the session; they replace the homework planned for that day (10 to 20 minutes in total).

CQFD, exercise 12 p. 156, questions 1 and 2. Calculate the difference, study its sign and translate this into the relative position of the curves.

CQFD, exercise 16 p. 156, questions f and g. For each function, give an interval on which it is both negative and increasing; justify the two properties separately.